

Governing Datacenter Growth: Regulatory Challenges, Technical Constraints, and Policy Responses Across States

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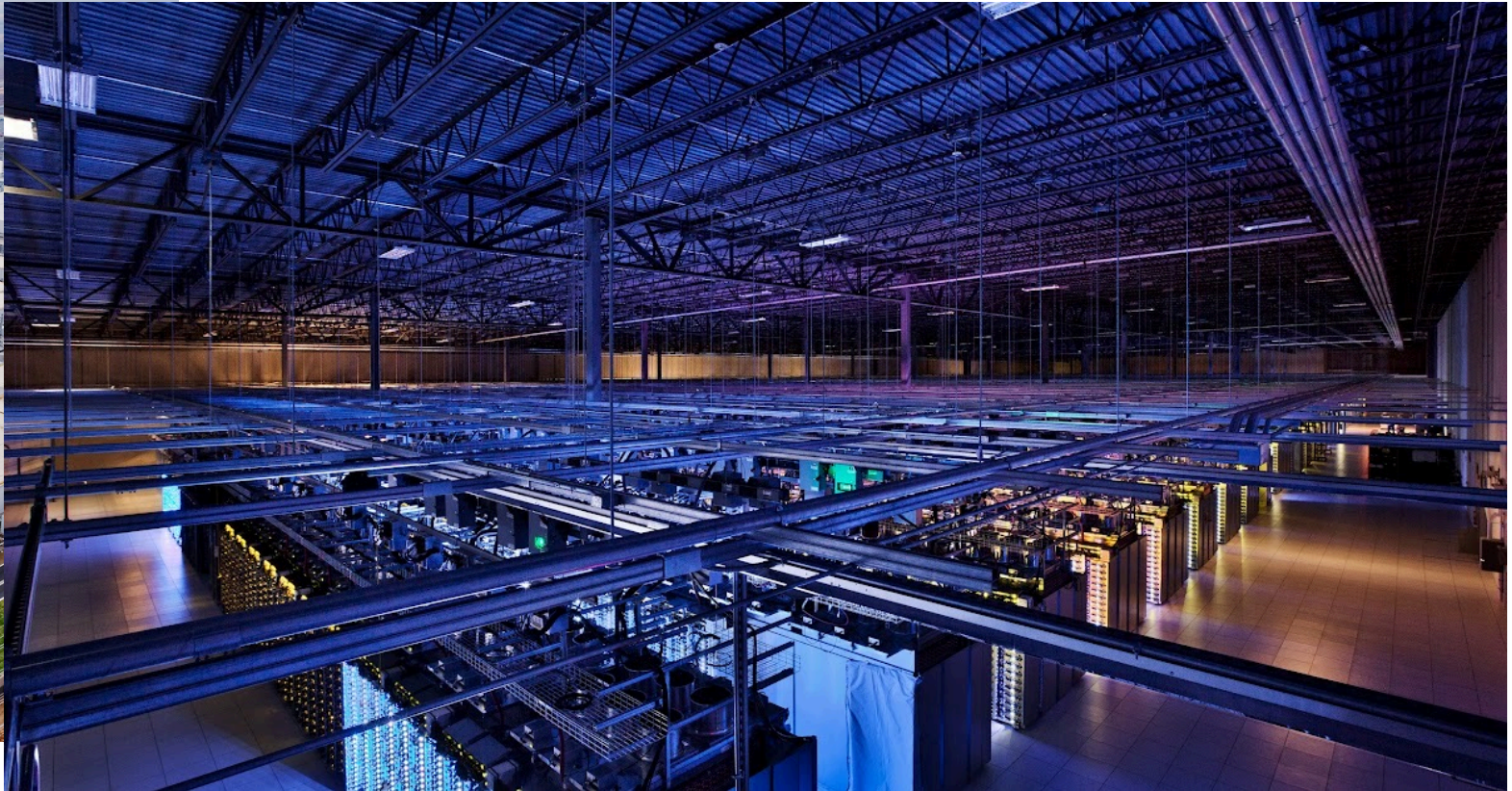
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Georgia Institute of Technology

What do datacenters actually look like?



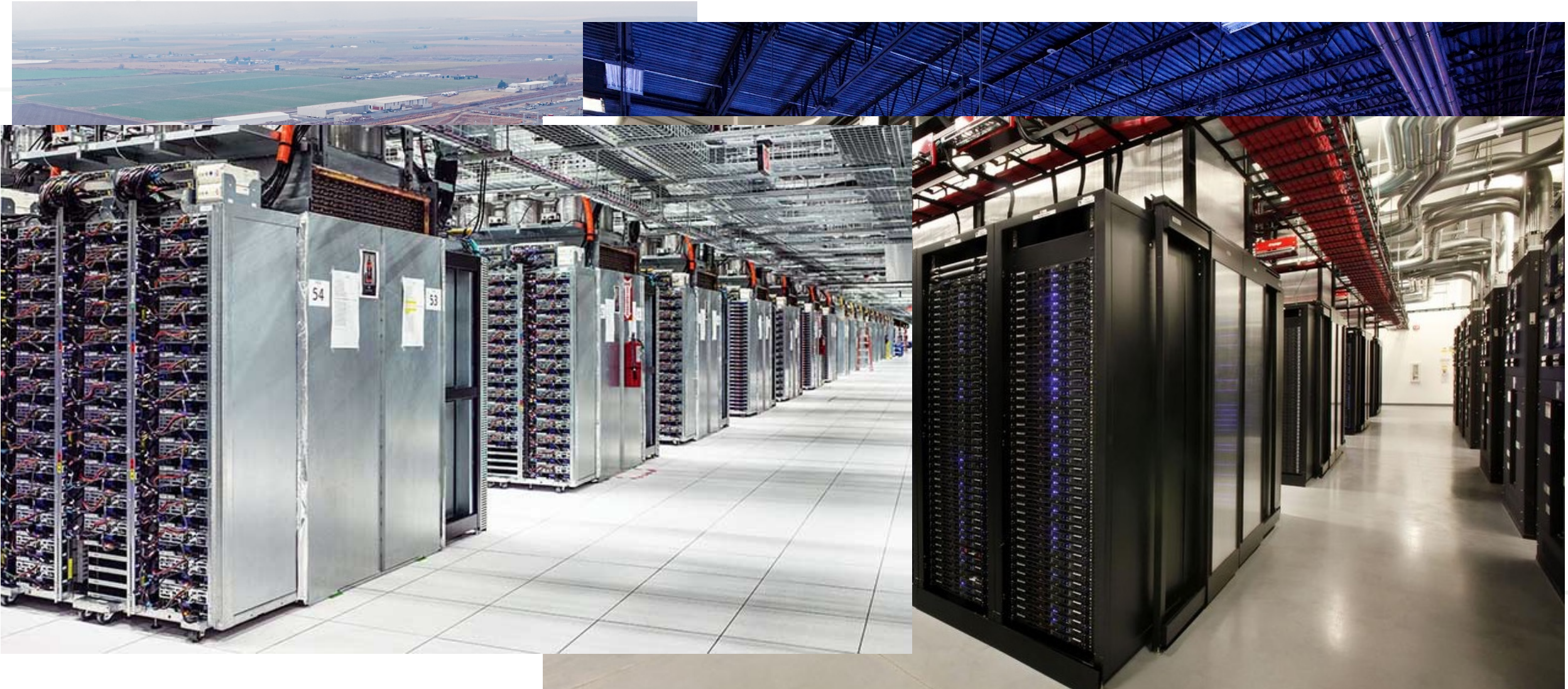
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each drawing up to 700 Watts**



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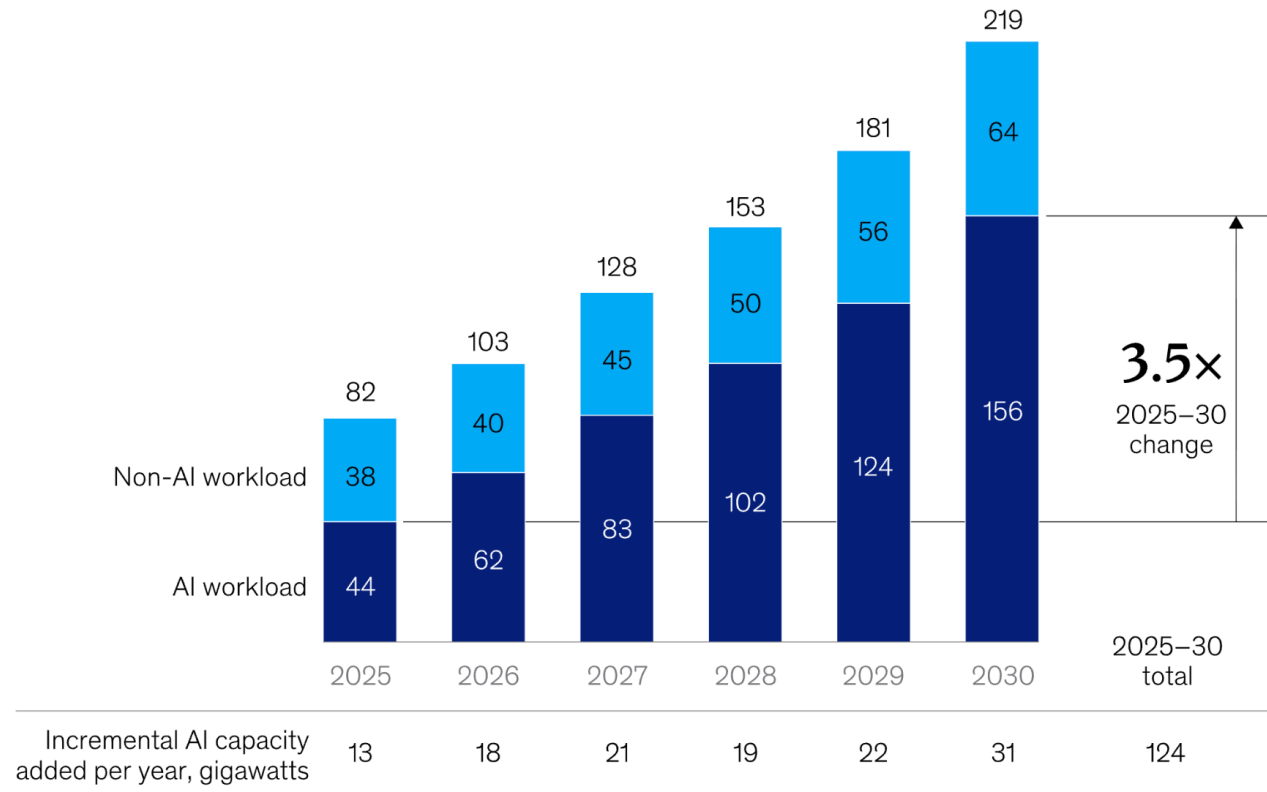
**8 Graphics Processing Units (GPUs)
each drawing up to 700 Watts**



Google and Nvidia plan to deliver a 1MW rack in the next couple of years.

Datacenter Applications

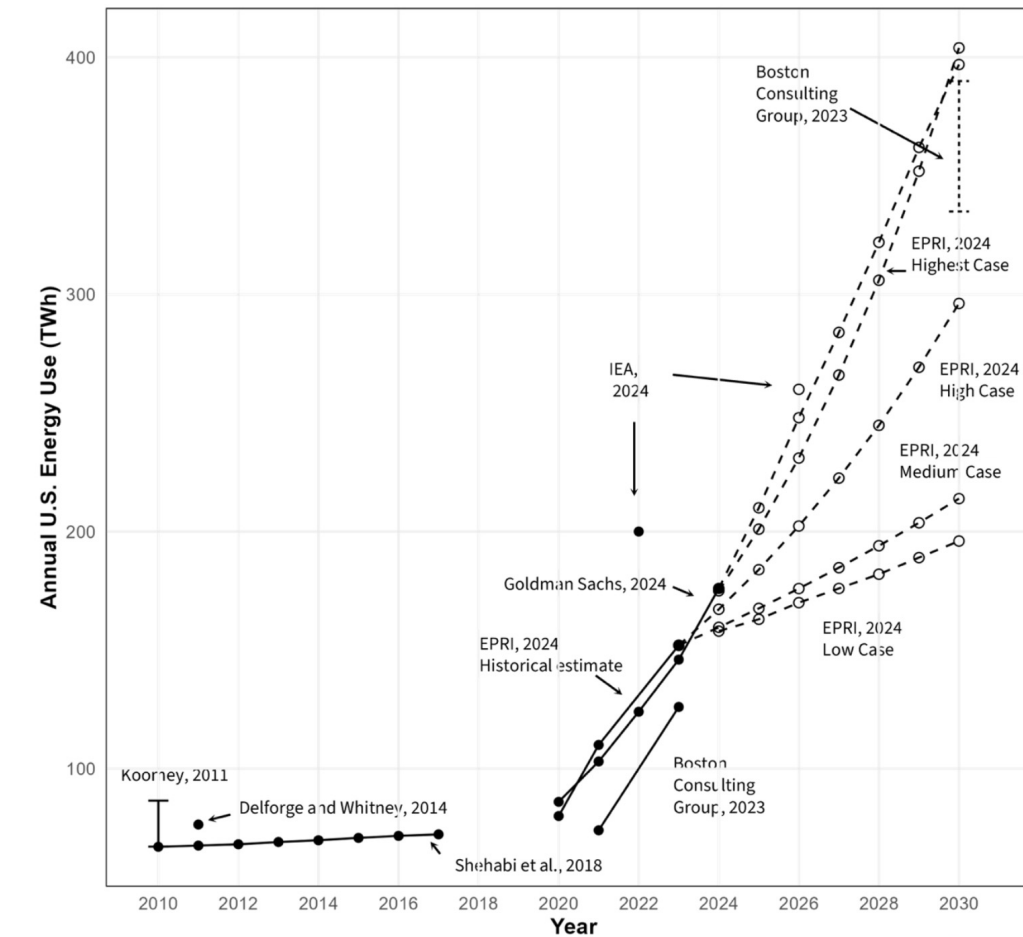
Estimated global data center capacity demand, 'continued momentum' scenario, gigawatts



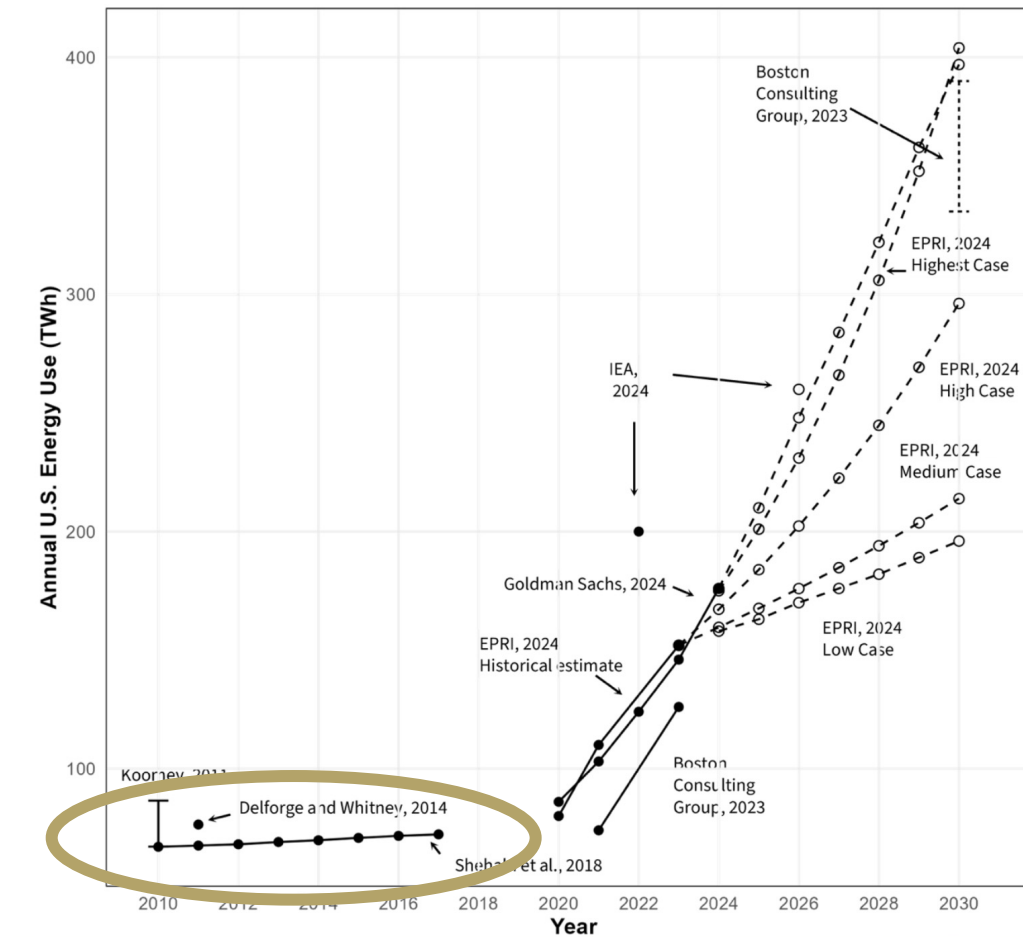
Note: Figures may not sum to totals, because of rounding.
Source: McKinsey Data Center Demand Model; Gartner reports; IDC reports; Nvidia capital markets reports

Datacenter Impacts

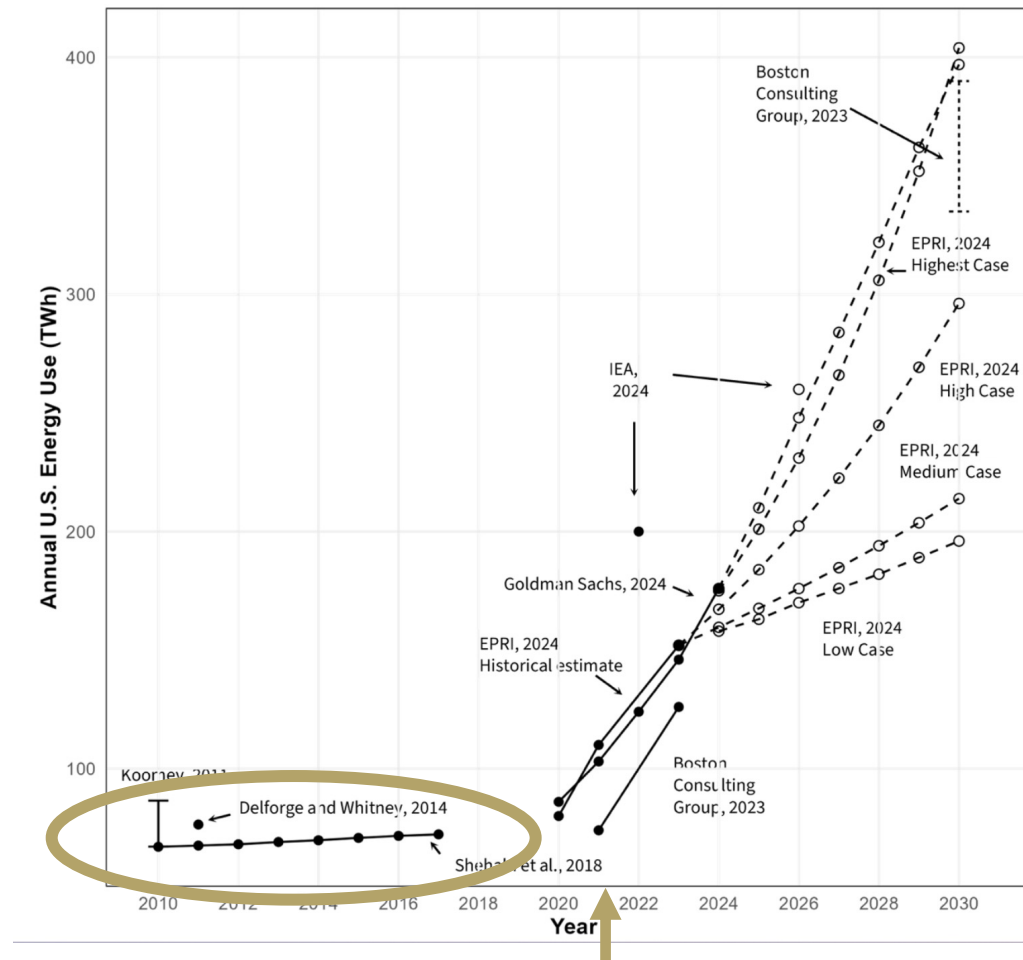
Datacenter Impact on the Grid



Datacenter Impact on the Grid



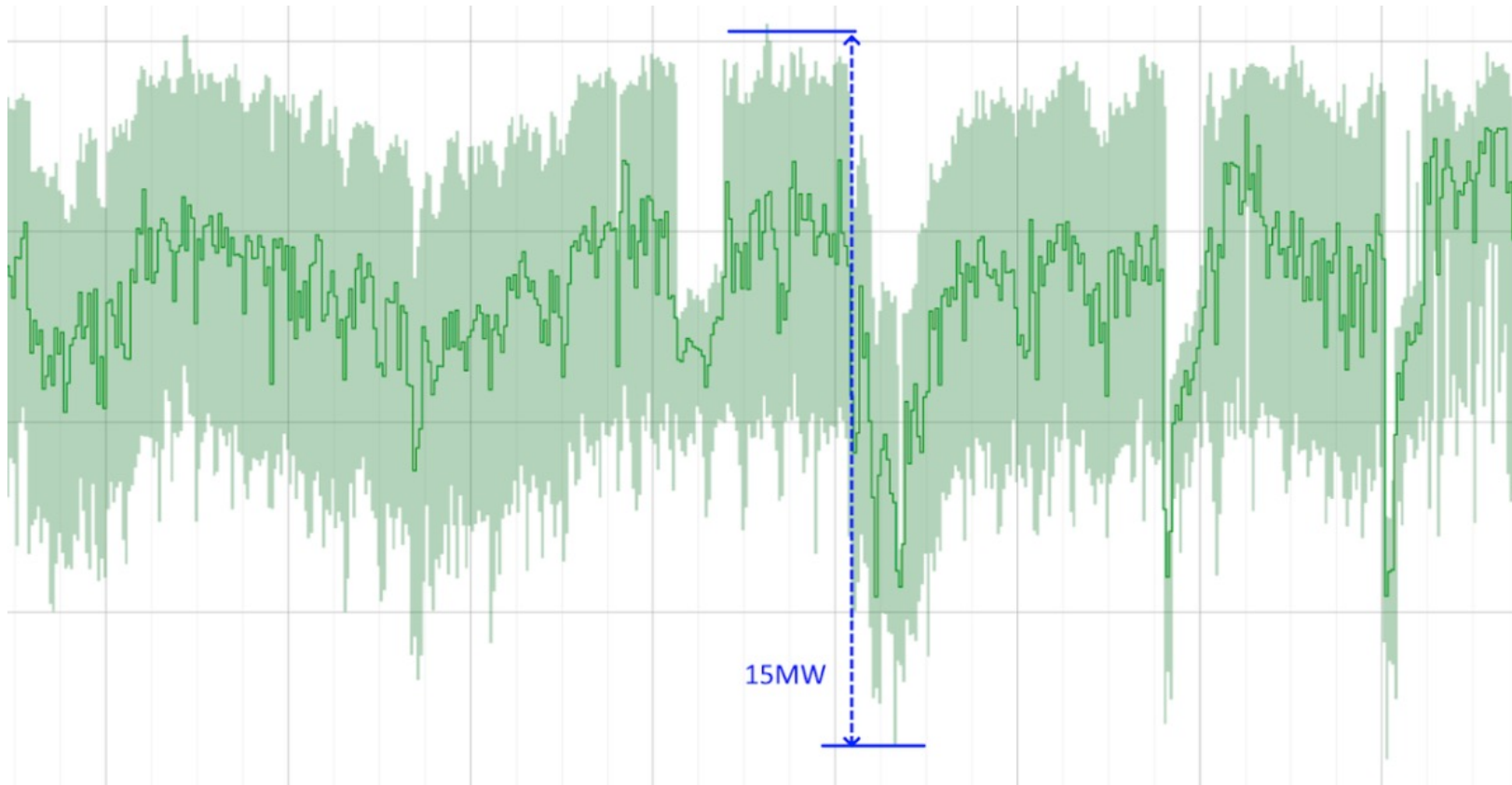
Datacenter Impact on the Grid



Training of GPT Model Started

Shehabi, A.; Newkirk, A.; Smith, S.; Hubbard, A.; Lei, N.; Siddik, M., et al. (2024). 2024 United States Data Center Energy Usage Report. Lawrence Berkeley National Laboratory. Report #: LBNL-2001637. <https://escholarship.org/uc/item/32d6m0d1>

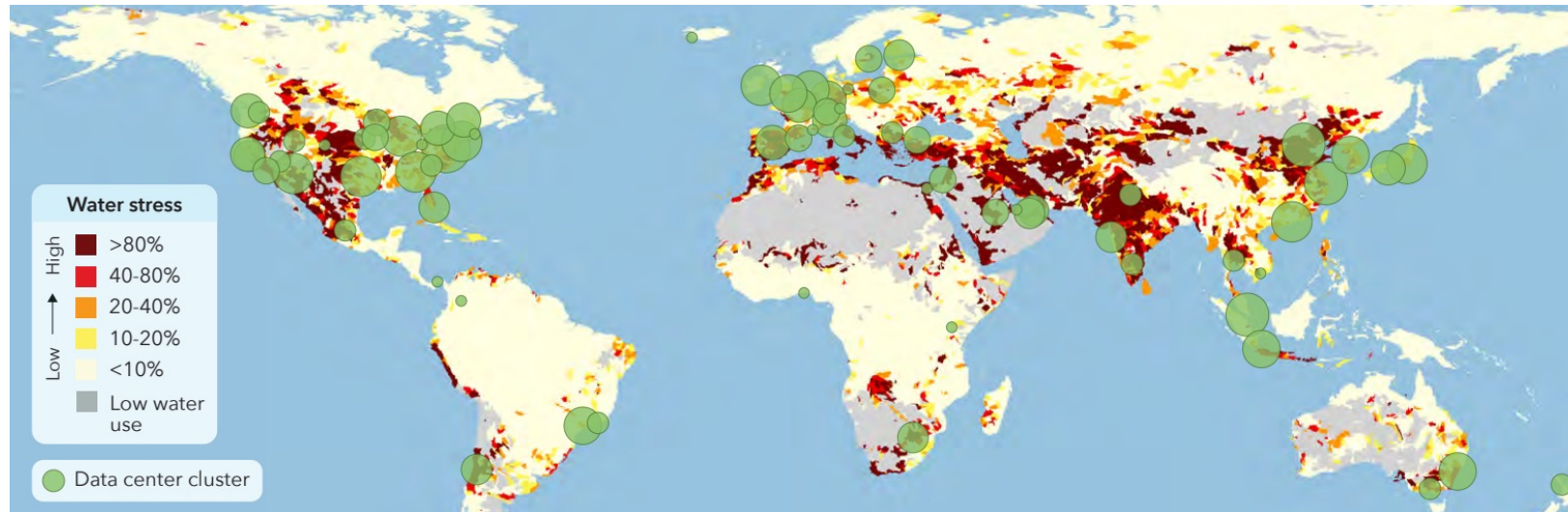
Datacenter Impact on the Grid



Swings equivalent to 12k households joining and leaving the grid in the span of seconds

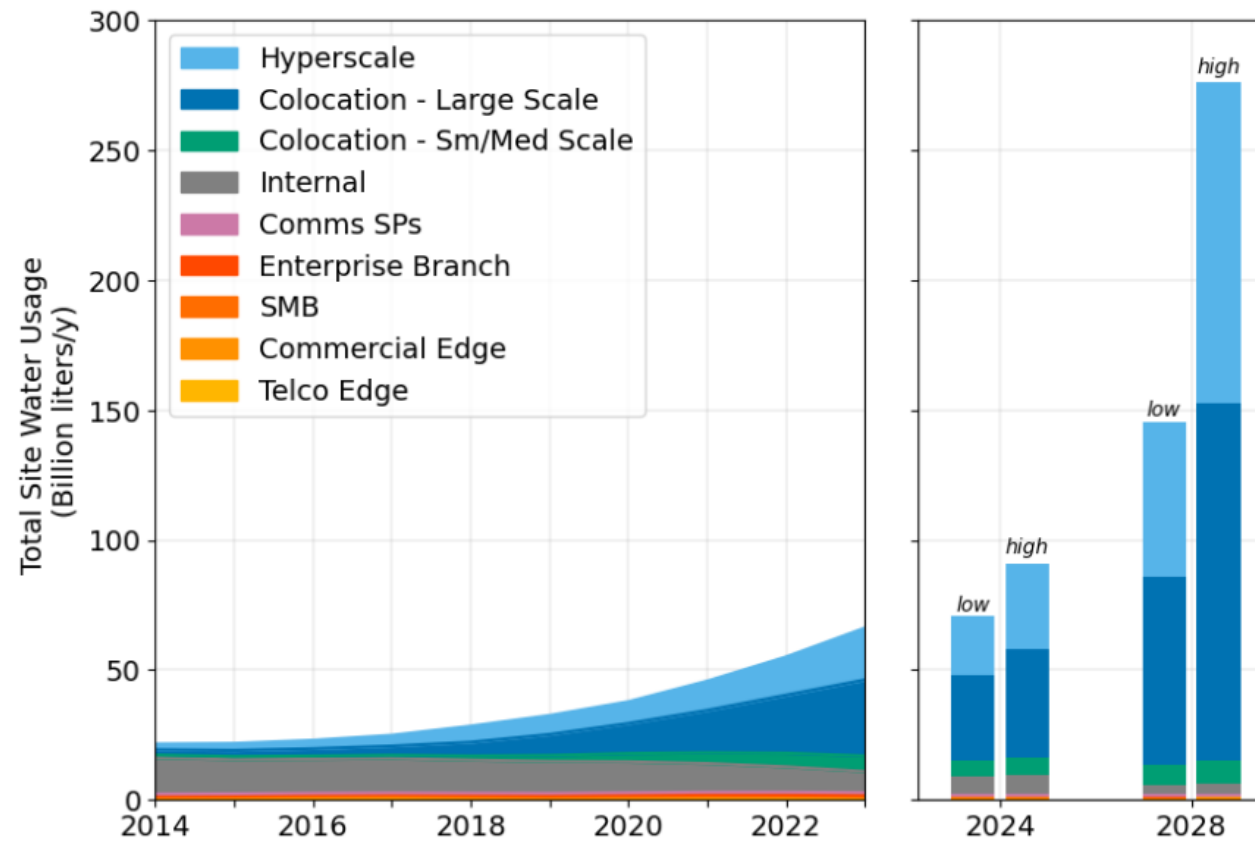
Datacenters require expansion of power generation and delivery capacities and pose risks on grid stability.

Datacenters and Water Stress



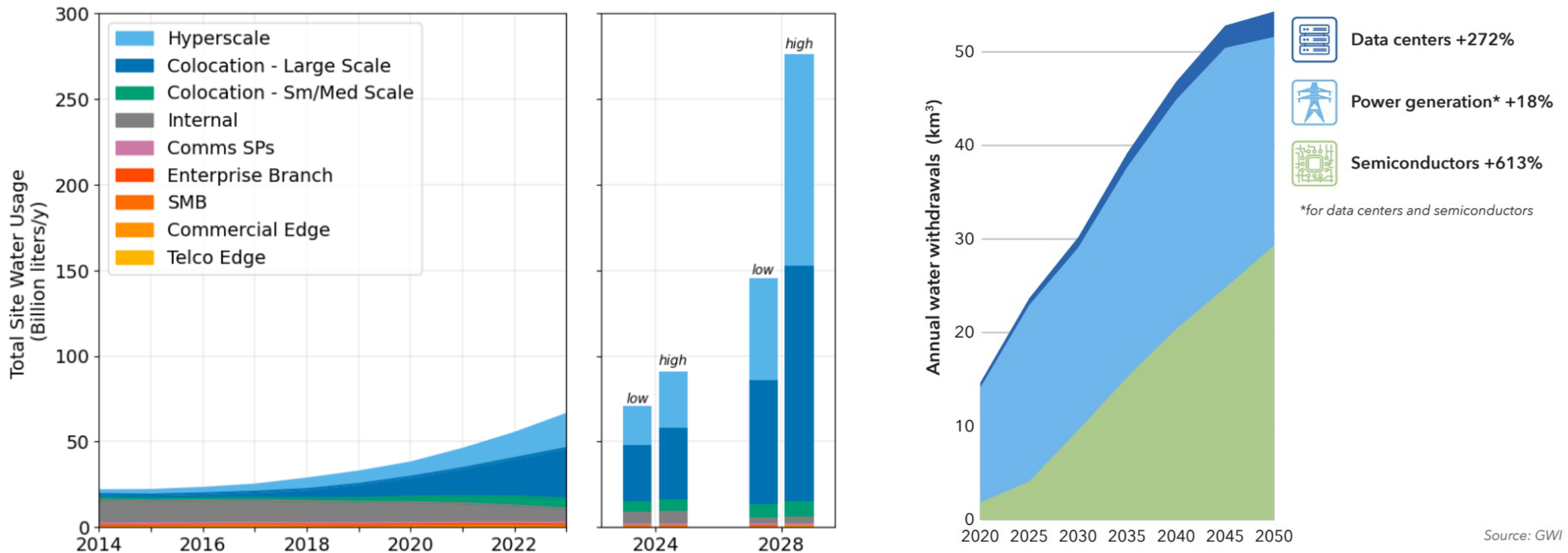
Datacenters are coming to areas with significant water stress

Datacenters and Water Stress



Predicted to consume hundreds of billions of liters per year

Datacenters and Water Stress



But context matters ...

Datacenters water use is alarming especially for water stressed regions, but their regulation should take their relative consumption into consideration.

Datacenter Regulation

Challenges

Long-lived commitments meet rapid development and changing workloads

- **Speed**

Development moves much faster than regulation and even infrastructure planning and upgrades

- **Scale**

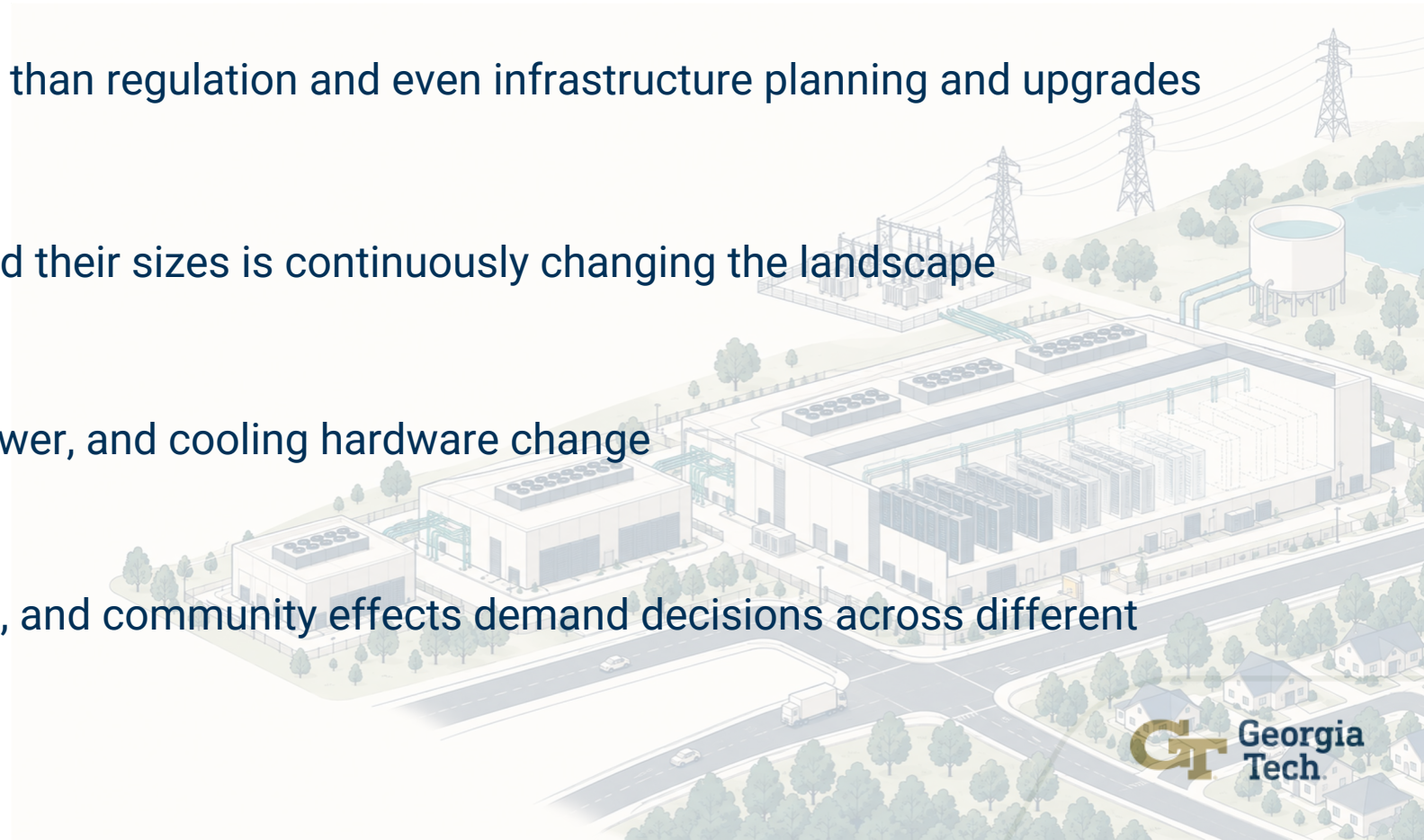
The number of proposed sites and their sizes is continuously changing the landscape

- **Turnover**

Buildings endure but compute, power, and cooling hardware change

- **Scope**

Power, water, land use, emissions, and community effects demand decisions across different policy areas.



Opportunities

Operators have unusual control over where and when they compute and how they power and cool it

- **Workload flexibility**

Shift suitable computing tasks across time and locations to ease grid constraints (demand-response) and make better use of available clean power

- **Energy resources**

On-site generation and energy storage can reduce demand on the grid or even become grid assets

- **Strategic siting**

Choosing sites that optimize for operator and community needs

- **Financial capacity**

Fund new clean generation, grid upgrades, water infrastructure, and community benefits through specific, long-term commitments



No single regulator sees that whole project

- Local

Land use, local water, noise, property taxes, and community impacts

- State

Utility regulation and costs, water policy, incentives

- Federal

Transmission, wholesale markets, environmental and reliability rules

Our Focus: State-level Legislation

States have broad control over utilities, resources, and tax incentives and offer transferable regulatory models, while local decisions are too case-specific.

Five states, different emphases

- **Texas**

Grid reliability: secure generation capacity, transparently track consumption, protect rate payers, and require emergency flexibility.

- **Minnesota and Virginia**

Integrated resource regulation: power, water, and emissions.

- **South Dakota**

Ratepayer and community protection: attributable-cost recovery, public water reporting, and preserve local authority.

- **New York**

Precautionary review and ambitious goals

A Review of Data Center Policies along Seven Dimensions

C1. Definition

C2. Grid + interconnection

Project-readiness screening, study and queue rules, load forecasting, transmission planning, and reliability conditions.

C3. Flexibility

Defined ability to shift or reduce demand, respond to emergencies, and measure delivery against operating commitments.

C4. Costs + transparency

Who pays for upgrades and stranded assets, plus whether electricity use, costs, and obligations are measured and disclosed.

C5. Clean energy

Incremental clean-capacity procurement, efficiency or sustainable-design requirements, and controls on on-site emissions.

C6. Water

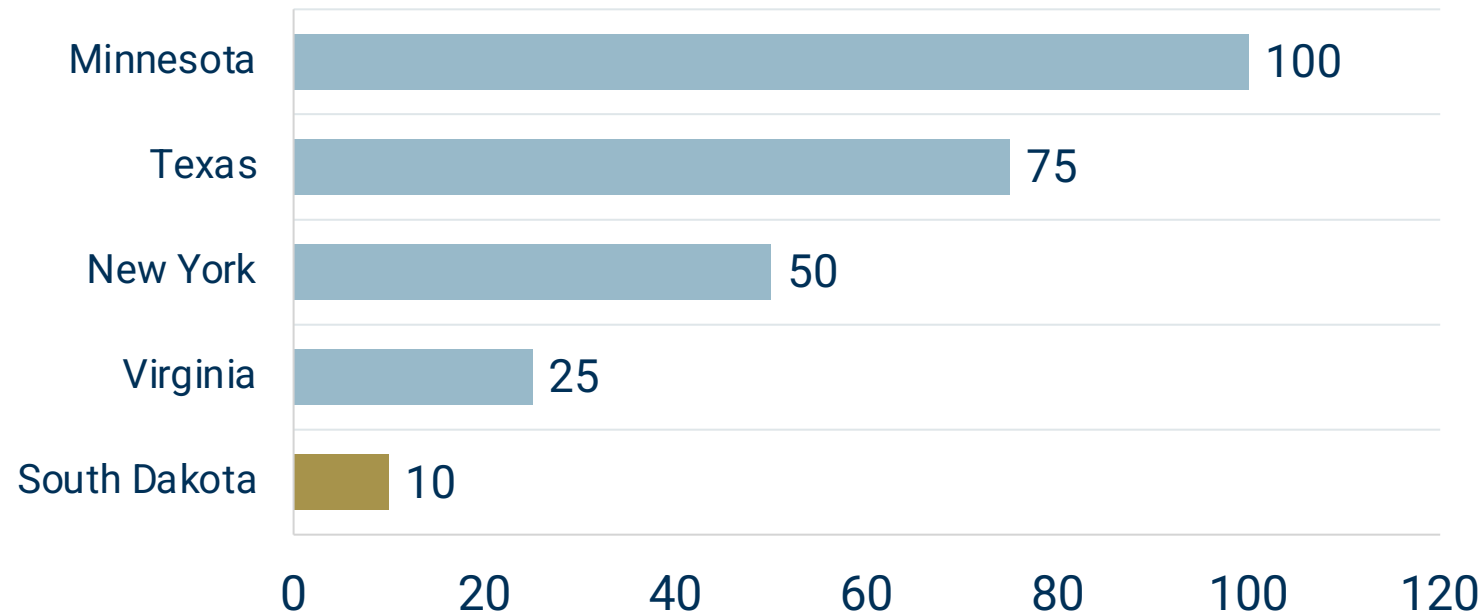
Advance supply review, permits, cooling choices, conservation and reuse, public reporting, and authority to limit use.

C7. Implementation

Siting and local authority, cumulative-impact analysis, public participation, and disclosure of community effects.

C1. Definition of a datacenter

Principal electric-demand triggers in the paper, in MW



20 MW

Illustrative facility

Within South Dakota's
data-center definition

A lower trigger brings more facilities into the principal regime

C2. Texas screens project readiness

SB 6 links credible demand forecasts to customer commitments



Credible projects reduce the risk of stranded grid investments

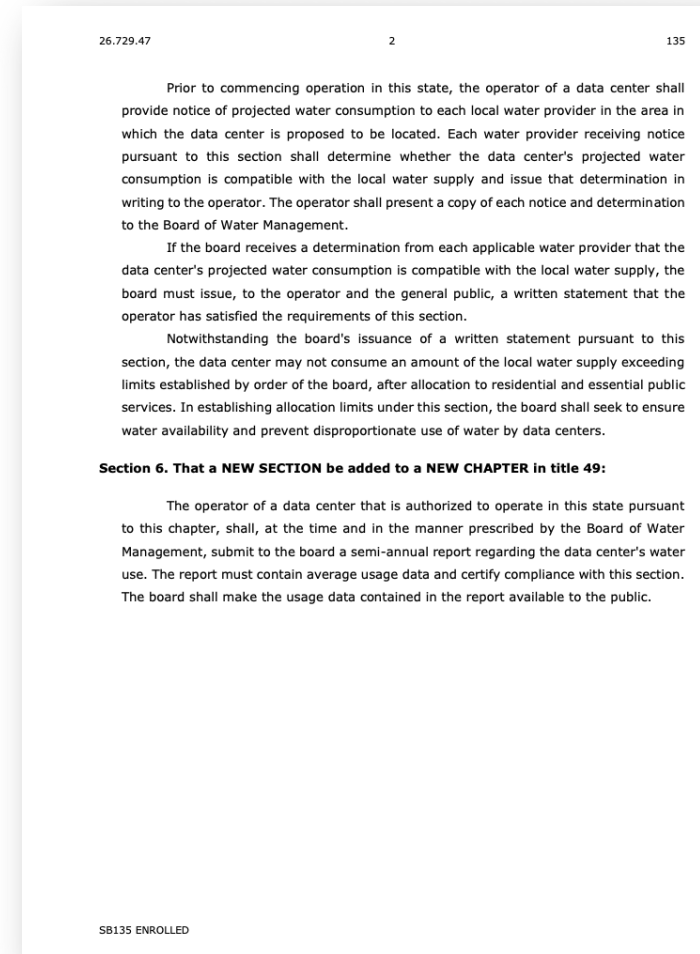
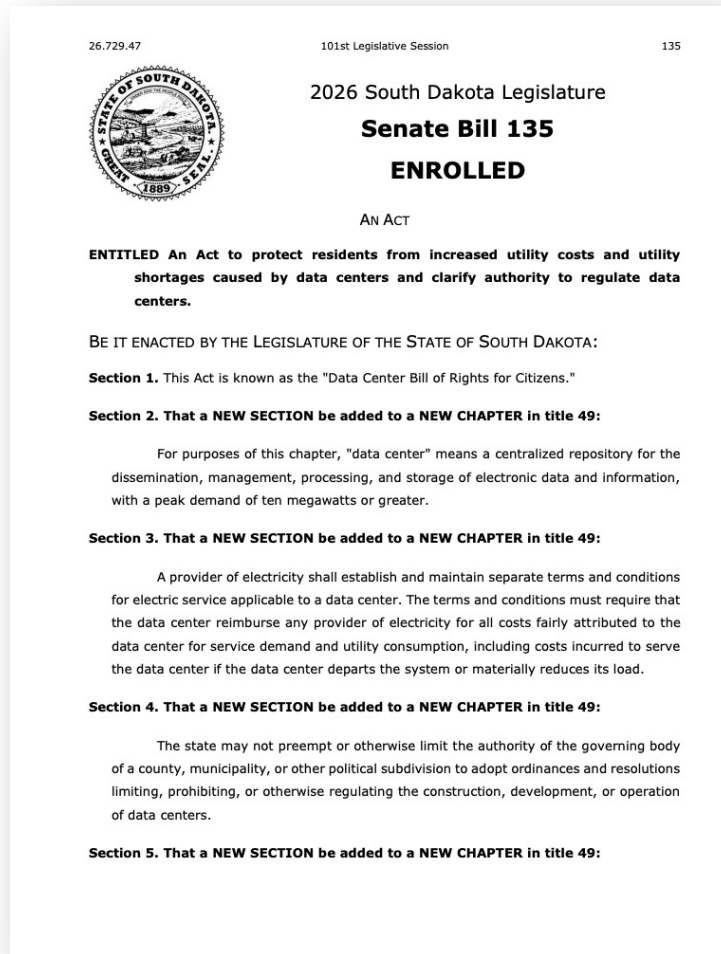
C3. Texas makes emergency flexibility enforceable

The key distinction is who can call on the load, and under what conditions

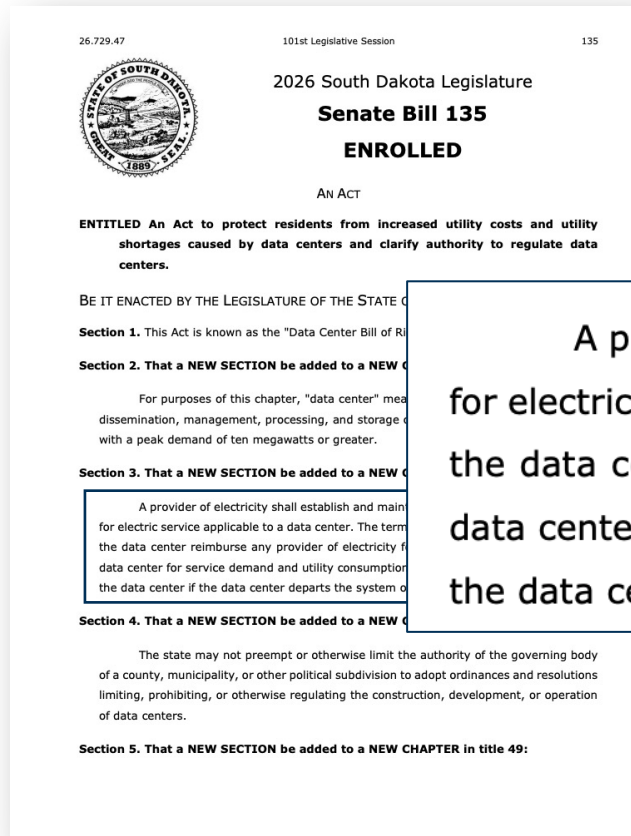
	Texas	Virginia
Legal mechanism	Emergency direction	Voluntary program
Facility response	Deploy qualifying backup or curtail grid demand	Shift or reduce use under program arrangements
Constraint	Environmental limits still apply	Participation and terms depend on program design

A potential capability becomes useful when the operating terms are clear

C4. South Dakota emphasizes consumer protections



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A provider of electricity shall establish and maintain separate terms and conditions for electric service applicable to a data center. The terms and conditions must require that the data center reimburse any provider of electricity for all costs fairly attributed to the data center for service demand and utility consumption, including costs incurred to serve the data center if the data center departs the system or materially reduces its load.

Specify both who bears the risk and what information makes that responsibility verifiable.

C5. Two paths for greener datacenters in Minnesota

Procurement and incentive conditions create different obligations

	Clean energy and capacity tariff	Certification for incentives
Participation	Optional	Required for qualifying incentive recipients
Obligation	Fund proportional costs of new clean energy and capacity	Obtain approved design or energy-management certification
Consequence	Customer pays for the resources it contracts for	Failure can trigger tax recapture

Service arrangements and incentive conditions give the state different ways to influence energy choices.

C6. Minnesota examines cooling before permitting

Special review: more than 100 million gallons/year of consumptive use



Water availability

Cooling and reuse

Other users and ecosystems

The trigger also requires a new or amended appropriation permit

C7. Binding law still needs implementation

Texas illustrates why enactment and operational completeness are different

	Binding foundation	Details still developing
Texas	Grid-screening and emergency duties	Interconnection procedures, reliability-service design
New York	Temporary restraint on covered permits	Permanent environmental and operating standards

Legal implementation and measured policy outcomes are yet to be determined.

States Offer Complementary Lessons

Criterion	Texas	Minnesota	South Dakota	New York	Virginia
Grid + interconnection	Leading	Moderate	Moderate	Emerging	Strong
Flexibility	Leading	Moderate	Limited	Emerging	Strong
Costs + transparency	Strong	Strong	Strong	Emerging	Leading
Clean energy	Limited	Leading	Limited	Emerging	Moderate
Water	Limited	Leading	Strong	Emerging	Strong
Community review	Limited	Strong	Moderate	Leading / interim	Strong

Build Assets, Prevent Liabilities

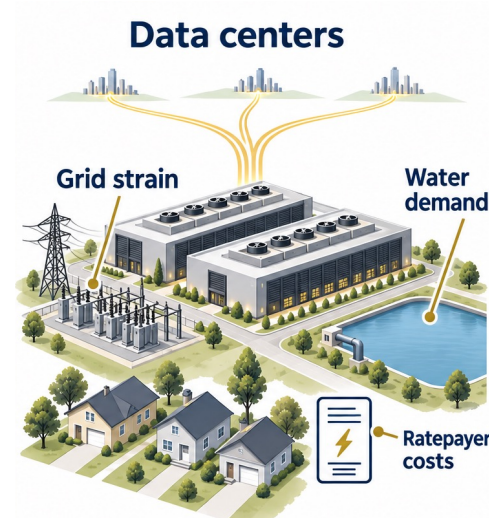


Physical Connectivity

The 20th-century model for national growth focused on the interstate system to enable commerce and security.

Federal Mandate: Critical for interstate commerce and national defense.

Local Burden: Noise, land-use disruption, and pollution concentrated in communities.



Digital Connectivity

The 21st-century equivalent, where data centers serve as the backbone for AI, security, and global digital trade.

Federal Mandate: Essential for AI supremacy and digital economic sovereignty.

Local Burden: Energy grid strain, water extraction, and ratepayer cost-shifting.

Build Assets, Prevent Liabilities

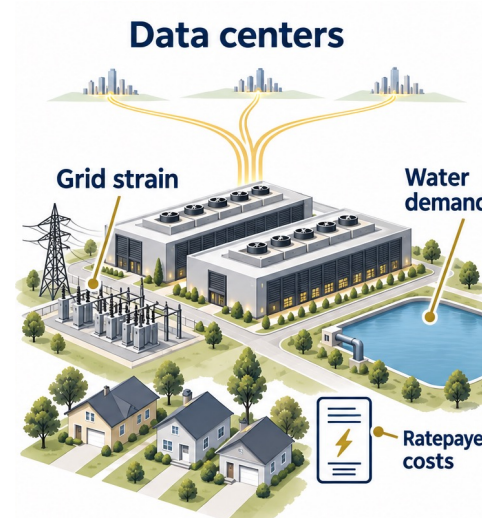


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Questions?